2024

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-1016

(Real Time System)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1** and **2** and ***any four*** from the rest.

1. Answer the following questions (***any five***) : 2×5

(a) What are 'context switching' and 'self-suspension'? What are their implications for real-time system design?

(b) "The worst-case response time of a task occurs when it is out of phase with other higher priority tasks."— Justify the statement.

(c) "RMA is preferred over the time-sliced round-robin algorithm for scheduling a set of soft real-time tasks on a uni-processor system."— Justify the statement.

(d) You need to track patient vitals and trigger alerts if vital signs exceed safe time limits. Write a suitable query to retrieve critical vitals :

Table : `patient_vitals` (patient_id, heart_rate, blood_pressure, last_updated).

(e) Can the basic priority ceiling protocol be used with dynamic priority scheduling? Would it be a good idea to do so? Explain your answer.

(f) What do you mean by the "fail-safe" state of the system? What is the statement's implication : "Safety-critical real-time systems do not have a fail-safe state."

(g) What do you understand by exception handling?

2. Answer the following questions (***any five***) : 4×5

(a) Critically Comment on the desirable characteristics of a real-time system.

(b) What do you mean by the scheduling point of a task scheduling algorithm? How are scheduling points determined in clock-driven, event-driven, and hybrid schedulers?

(c) Critically comment on necessary and sufficient schedulability tests for a fixed priority system. What are the practical factors whose effect needs to be considered for such a schedulability test?

(d) What are the desirable features of any real-time operating system? Why is the standard version of LINUX not considered a real-time operating system?

(e) Are the techniques developed for real-time operating systems schedulers directly applicable to real-time database schedulers? Why or why not?

(f) What do you mean by fault tolerance? Compare and contrast hardware and software fault tolerance.

(g) What do you mean by frame size for a set of periodic tasks? What are the constraints to be satisfied to get a suitable frame size for a given set of periodic tasks?

3. (a) Describe how each of the following types of server algorithms works (in terms of release, replenishment and consumption), assuming that Ts = (ϕs, Ps, Es , Ds) :

(i) deferrable server

(ii) sporadic server

(b) Justify the requirement of acceptance test to schedule sporadic jobs along with periodic and aperiodic jobs. (4+4)+2

4. (a) What do you mean by optimality of RM and DM algorithms?

(b) Using a cyclic real-time scheduler, suggest a suitable frame size that can be used to schedule three periodic tasks, Tl, T2, and T3, with the following characteristics :

Task	Execution Time (ms)	Relative Deadline (ms)	Period(ms)
Tl	20	100	100
T2	20	80	80
T3	30	150	150

4+6

5. (a) For the following set of jobs with associated parameters, find the job schedule using the priority ceiling protocol.

Job	Release Time	Execution Time	Priority	Critical Section
J1	7	3	1	[Shaded;1]
J2	5	3	2	[black;1]
J3	4	2	3	
J4	2	6	4	[Shaded;4[Black;1.5]]
J5	0	6	5	[Black;4]

(b) Explain how PCP is able to avoid deadlock, unbounded priority inversions, and chain blocking. 7+3

6. (a) What do you mean by 'Priority Inversion'? When does it occur?

(b) Two options for avoiding priority inversion are the basic priority inheritance protocol and the basic priority ceiling protocol. Discuss how these two protocols differ in terms of their scheduling, resource allocation, and priority inheritance rules.

(c) Explain why the basic priority ceiling protocol is immune to deadlock. 2+6+2

7. (a) How are the ACID properties of general-purpose DBMS compromised in RDBMS? Justify the deviations.

(b) In the context of IoT (Internet of Things), compare and contrast the commonly used communication protocols to enable communication between devices and systems.

(c) Write a short note with technical specifications on any micro-controller board. 3+3+4

8. (a) Explain the terms with reference to a set of periodic tasks : 'Critical Instants' and 'Busy Interval.'

(b) What is 'jitter', and what is its significance in the real-time system?

(c) What are the roles of sensors, actuators, and controllers in real-time systems? Give examples of these from practical systems. Give the general structure of a real-time system using a schematic diagram, clearly showing the controller, actuators, and sensors. 4+2+4